WORKSHEET 8.6

Regions with More Than One Piece

Unit 8: Applications of Integration

Key ideas

- If the graphs cross inside the interval, the top graph changes. Split the integral at each crossing.
- In one formula: area $= \displaystyle\int_a^b |f(x) - g(x)|\,dx$. On a calculator, enter the absolute value directly.
- $\displaystyle\int_a^b (f(x) - g(x))\,dx$ without the absolute value gives a **net** amount, in which pieces cancel.
- A region with a corner in its boundary also needs two integrals, one for each boundary curve.

Worked examples

Example 1 · A cubic and a line

Find the total area enclosed by $y = x^3$ and $y = 4x$.

Solution. They meet at $x = -2, 0, 2$. By symmetry, $2\displaystyle\int_0^2 (4x - x^3)\,dx$.

$2(8 - 4) = 8$.

Example 2 · Net against total

Compare $\displaystyle\int_0^{\pi} \cos x\,dx$ with the area between $y = \cos x$ and the x-axis for $0 \le x \le \pi$.

Solution. The integral is 0. The area is $\int_0^{\pi/2} \cos x\,dx + \left|\int_{\pi/2}^{\pi} \cos x\,dx\right|$.

The integral is 0; the area is 2.

Example 3 · Splitting at a zero

Find the area between $y = x^2 - 4$ and the x-axis for $0 \le x \le 3$.

Solution. $\displaystyle\int_0^2 (4 - x^2)\,dx = \frac{16}{3}$ and $\displaystyle\int_2^3 (x^2 - 4)\,dx = \frac{7}{3}$.

$\dfrac{23}{3}$.

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Practice

CORE

1. Find the area between $y = x - 1$ and the x-axis for $0 \leq x \leq 4$.

2. At what value of x in $(0, 3)$ do the graphs of $y = x^2$ and $y = 2x$ cross?

3. Find the area between $y = x^2$ and $y = 2x$ for $0 \leq x \leq 3$.

4. Evaluate $\int_0^3 (x^2 - 2x)\,dx$, and explain the result using Question 3.

EXAM LEVEL

5. Find the total area between $y = x^3 - x$ and the x-axis for $-1 \leq x \leq 1$.

6. Find the area between $y = \sin x$ and $y = \frac{1}{2}$ for $0 \leq x \leq \frac{\pi}{2}$.

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7. Find the area of the region in the first quadrant bounded by $y = x^2$, $y = 2 - x$ and the x-axis.

8. *Calculator.* Find the area between $y = \cos x$ and $y = \dfrac{x}{2}$ for $0 \le x \le 3$.

Question 9 · Free-response practice

Let $f(x) = x^3 - 5x^2 + 4x$.

(a) Evaluate $\displaystyle\int_0^4 f(x)\,dx$.

(b) Find the total area between the graph of f and the x-axis for $0 \le x \le 4$.

(c) Explain why the answers to parts (a) and (b) differ, and write a single integral that gives the area in part (b).

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Worked solutions

1. $\frac{1}{2} + \frac{9}{2} = 5$.
2. $x = 2$.
3. $\int_0^2 (2x - x^2)\,dx + \int_2^3 (x^2 - 2x)\,dx = \frac{4}{3} + \frac{4}{3} = \frac{8}{3}$.
4. 0: the two pieces have equal area, and they cancel.
5. $2\left(\frac{1}{4}\right) = \frac{1}{2}$.
6. They cross at $x = \frac{\pi}{6}$: $\left(\frac{\pi}{12} - 1 + \frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6}\right) = \sqrt{3} - 1 - \frac{\pi}{12}$.
7. The curves meet at $x = 1$: $\int_0^1 x^2\,dx + \int_1^2 (2 - x)\,dx = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$.
8. $\int_0^3 \left|\cos x - \frac{x}{2}\right|\,dx \approx 3.293$ (the graphs cross at $x \approx 1.030$).
9. (a) $\left[\frac{x^4}{4} - \frac{5x^3}{3} + 2x^2\right]_0^4 = -\frac{32}{3}$.

   (b) $f(x) = x(x-1)(x-4)$: $\int_0^1 f(x)\,dx = \frac{7}{12}$ and $\int_1^4 f(x)\,dx = -\frac{45}{4}$. Area $= \frac{7}{12} + \frac{45}{4} = \frac{71}{6}$.

   (c) Part (a) counts the piece below the axis as negative, so the pieces partly cancel. Area $= \int_0^4 |f(x)|\,dx$.

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